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The Selling closure: a complete operator set for reindexing, and the price of pseudo-symmetry

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The reduced cells of a lattice and the integer matrices between them form a groupoid, the Selling closure: the matrices from a reduced cell to itself are the lattice point group, those between two reduced cells form one coset of it, and a reindexing is an arrow between two reduced cells. Every such matrix, for any lattice, lies in one fixed set of $6096 = 48 \times 127$ matrices with entries up to ± 3, by a short lemma and an exhaustive enumeration, machine-checked in Lean; the arrows depend only on which conorms of the reduced cell vanish. For a measured cell the reduced cells within a strain tolerance are finite and are enumerated exactly, so reindexing returns every proper operator within a stated price band, with a certificate of when the fixed set alone suffices. Prices are displacements under the action of the unimodular group on metric tensors: the candidates lie between two cosets of the near-symmetries, these are bounded by the symmetries of the nearest symmetric lattice, and a pseudo-twofold costs exactly twice its distance to the metrics it would leave invariant, about 5% per degree of Le Page obliquity. In the PDB, the cell alone cannot exclude an alternative indexing from a higher lattice for 7.5–8.6% of entries, and ranks a wrong indexing first for about 0.1–1.2% of pairs of one crystal form.

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1. Introduction

Take a primitive monoclinic lattice with $a = 30$, $b = 40$, $c = 50$ Å and $\beta = 100°$. Its Selling-reduced cell has two right angles between vectors of the reduced superbase, and a measured cell lies slightly to one side or the other of each. One data set therefore reduces to the cell 30, 40, 50 Å, $\beta = 100°$, and another, equally correctly, to 30, 40, 53.7 Å, $\beta = 113.4°$, built on a and $a + c$ (Section 3.2). The matrix between the two is not a symmetry of the lattice. A program that reports the Niggli cell meets the same multiplicity in another form: the cell on a and $a + c$ is never a Niggli cell, but at each right angle the Niggli cell changes between its two types, with three acute or no acute angles (Gruber, 1973; Andrews *et al.*, 1980), and two measurements of one crystal can be reported in Niggli cells related by a matrix that is not a symmetry either. Data-processing programs find such matrices by trying a list: small integer matrices between the two reduced cells, the matrices that unit-cell similarity searches generate (Grosse-Kunstleve *et al.*, 2002), or the operations of the lattice group found by a symmetry search (Le Page, 1982; Sauter *et al.*, 2004). On ordinary cases these searches succeed. This paper asks which integer matrices can relate two reduced cells, how to enumerate every candidate within a stated tolerance, and how to price the pseudo-symmetric alternatives that a search returns.

The multiplicity has a classical description. A superbase is four lattice vectors summing to zero; it is obtuse, or Selling-reduced, when all six pairwise dot products are non-positive (Selling, 1874; Conway & Sloane, 1992), and the obtuse superbase is unique except when a dot product vanishes (Delaunay, 1933). Kurlin (2022) listed all obtuse superbases of each Voronoi type. Delone's 24 sorts classify lattices by the zeros and equalities among the six dot products (Delone *et al.*, 1975; Andrews & Bernstein, 2023). The S^6 representation treats the six numbers as a point of a six-dimensional space, with boundary transformations between the reduced presentations of a lattice, and underlies the distances CS6Dist and NCDist and the Bravais-type determination of SELLA (Andrews *et al.*, 2019*b*, 2023; Andrews & Bernstein, 2014); that a boundary transformation at a zero component carries one reduced presentation to another is noted there (Andrews *et al.*, 2023, Section 6.3). The present paper does not work in S^6: it works with the reduced superbases themselves, the integer matrices between them, and distances between metric tensors that are the same in every basis. Oishi-Tomiyasu (2012, 2024) gives error-stable Bravais-lattice determination in Selling coordinates with a provably sufficient list of basis transforms. The instability of the Niggli cell (Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same multiplicity seen from the Niggli side, and the operators between two settings of one lattice form a coset of its symmetries (Flack, 1987).

This paper adds four things, all of them theory with a guarantee attached.

1. *The closure as a groupoid.* The obtuse superbases of a lattice in all their labellings, with the unimodular matrices between them, form a groupoid: its loops are the point group, its arrows between two reduced cells a coset, and the symmetries split into those that relabel a reduced cell and those that carry it to another (Section 3.1).

2. *A complete operator set.* Every arrow of every closure

lies in one set $\mathcal{U}$ of $6096 = 48 \times 127$ matrices, with entries up to ± 3; the arrows depend only on the zero pattern of the reduced cell. The proof is a short lemma and an exhaustive enumeration, machine-checked (Section 3.3, Lemma A2). Where the entries exceed 1 is stated exactly, and why reduction reaches those configurations near three right angles is proved (Section 5.1, Proposition A11).

3. *An exact enumeration under a tolerance.* For a measured cell, the superbases that are reduced for some metric within a strain η are finitely many and are enumerated exactly (Lemma A6), and reindexing a new cell against a reference widened this way returns every proper operator within a price band (Lemma A7). When the widened set lies in the first layer of $\mathcal{U}$, which the enumeration reports, the same candidates come from a lookup of the 6096 matrices between the two reduced cells (Remark A8), a fixed table any program can use.

4. *Prices as displacements.* A candidate is priced by the affine-invariant distance between the two metrics, the displacement of the action of $GL_3(\mathbb{Z})$. The candidates lie between two cosets of the near-symmetries, the near-symmetries are bounded by the symmetries of the nearest symmetric lattice (Proposition A10), and a pseudo-twofold costs exactly twice its distance to the metrics it would leave invariant (Proposition A9).

As an application, the price between deposited cells of one crystal form sets the scale of the tolerance, and comparing it with the prices of the pseudo-symmetries in the PDB bounds how often the cell alone cannot decide the indexing (Section 5.4). Everything is implemented in the open-source package `agentsg` (Zwart, 2026), which also reproduces every number in the paper. The searches in use find the true operator in ordinary cases, and nothing here is offered as a faster or more successful search; what is new is the guarantee, a statement for each cell and tolerance that the candidate set is complete, and a price for every alternative it contains.

Section 2 collects the classical definitions, Section 3 the theory with a table of what is proved, Section 4 the methods and tests, and Section 5 the results, ending with the PDB application; proofs are in Appendix A, and catalogues, further tables and the computational checks in the supporting information.

2. Background

2.1. Superbases, conorms and the metric tensor

Selling reduction works with four vectors: the cell vectors a, b, c and $d = -a - b - c$. The four form a *superbase*, and any three of them are a basis of the lattice. The six negated pair products

$$p_{da} = -d \cdot a, \quad p_{db} = -d \cdot b, \quad p_{dc} = -d \cdot c,$$

$$p_{ab} = -a \cdot b, \quad p_{ac} = -a \cdot c, \quad p_{bc} = -b \cdot c$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992). Conorm vectors are always written in this slot order, da, db, dc, ab, ac, bc; the Selling parameters of S^6 are the same numbers with the opposite sign, in another order. A superbase is *obtuse*, which is to say Selling-reduced, when all six conorms are non-negative. The metric tensor in the basis (a, b, c) is linear in the conorms,

$$\mathbf{G} = \begin{pmatrix} p_{ab} + p_{ac} + p_{da} & -p_{ab} & -p_{ac} \\ -p_{ab} & p_{ab} + p_{bc} + p_{db} & -p_{bc} \\ -p_{ac} & -p_{bc} & p_{ac} + p_{bc} + p_{dc} \end{pmatrix}, \tag{1}$$

so the conorms are six cell parameters. A zero conorm is a right angle between two of the four vectors; the set of slots that vanish is the *zero pattern*, and the set of conorm vectors with $p \geq 0$ and those slots zero is a *face* of the cone of non-negative conorms. A face describes lattices only if its non-zero slots connect the four vectors. Up to relabelling there are six kinds of face: no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector, and three zeros forming a triangle or a path. There are 38 faces in all. The triangle and the path describe the same lattices, those with three mutually perpendicular lattice vectors, read in different cells, so the six kinds of face are the five Voronoi types (Voronoi, 1908; Conway & Sloane, 1992). A *frame* is a choice of basis in which a metric tensor or a superbase is written. A *partial sum* of a superbase is the sum of some of its four vectors. Two lattice vectors are in the same *parity class* when their difference is twice a lattice vector; written as $x = n_1 a + n_2 b + n_3 c$, a vector's class is given by which of the integer coefficients n_i are even and which odd (Appendix A). There are eight classes, and the seven partial sums, up to sign, lie one in each class other than the doubled vectors.

2.2. Flips and swaps

Two kinds of integer operator act on a superbase.

Flips. A flip at the conorm p_{ab} replaces a by $-a$ and adds the old a to the two other vectors, $(a, b, c, d) \mapsto (-a,\ b,\ c + a,\ d + a)$. The new conorms are $p'_{ab} = -p_{ab}$, $p'_{ac} = p_{da} + p_{ab}$, $p'_{da} = p_{ac} + p_{ab}$, $p'_{bc} = p_{bc} + p_{ab}$, $p'_{db} = p_{db} + p_{ab}$ and $p'_{dc} = p_{dc} - p_{ab}$. Selling reduction applies flips at negative conorms until none remain. When p_{ab} is exactly zero the flip exchanges the slots ac and da and carries one obtuse superbase to another of the same lattice, which is why the reduced cell of a lattice with a right angle is not unique. In the basis (a, b, c) each flip is a fixed 3×3 integer matrix of determinant -1 that is its own inverse (Table 1). A word of flips such as $ab.bc.db$ is applied left to right, each flip being read in the labelling current at that step, and its matrix is $\mathbf{F}_{ab}\mathbf{F}_{bc}\mathbf{F}_{db}$, columns being the new basis vectors in the old basis.

Table 1
The six Selling flips as 3×3 integer matrices in the basis (a, b, c), $d = -a - b - c$; columns are the new basis vectors in the old basis. Each has determinant -1 and squares to the identity. For the flips at da, db and dc the vector d is negated and the new a, b, c are listed.

flip at	new basis	matrix
ab	$(-a,\ b,\ a+c)$	$\begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
ac	$(-a,\ a+b,\ c)$	$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
bc	$(a+b,\ -b,\ c)$	$\begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
da	$(a,\ -a-c,\ -a-b)$	$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$
db	$(-b-c,\ b,\ -a-b)$	$\begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$
dc	$(-b-c,\ -a-c,\ c)$	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$

Swaps. A superbase is an unordered set of four vectors up to a common sign, so one reduced cell can be written in $4! \times 2 = 48$ ways. The 48 reorderings, with or without a change of the common sign, are the swaps. They change which vectors are called a, b, c and nothing else. Negating a single basis vector is not a swap: it does not preserve $a + b + c + d = 0$.

In S^6. On the six Selling parameters a flip acts as a boundary transformation and a swap as one of the 24 reflections of S^6 (Andrews *et al.*, 2019*b*); here both are kept as integer matrices on the basis, which is what a reindexing needs.

2.3. Delone sorts and the root invariant

Delone (published as Delaunay, 1933) classified lattices into 24 sorts by the pattern of zeros and equalities among the reduced conorms. Each sort lies in one Bravais type, and some Bravais types comprise several sorts (Delone *et al.*, 1975; Burzlaff & Zimmermann, 1985, 2006). We use the symbols C1 to A3 and the patterns of Andrews & Bernstein (2023) and SELLA (Andrews *et al.*, 2023), read in the slot order of Section 2.1; the correspondence with their slot order is given in the supporting information. The sorts are the Selling counterpart of the 44 Niggli lattice characters (Niggli, 1928; de Wolff & Gruber, 1991). They split a Bravais type by the shape of the reduced cell, not by its symmetry. A zero conorm restricts the metric without adding a symmetry: one right angle between two superbase vectors gives no twofold, so the triclinic sorts A2 and A3, with one and two zeros, are as triclinic as A1.

All obtuse superbases of a lattice carry the same six conorm values as a multiset, since a flip at a zero conorm only exchanges two slots. Which slots hold which value is not invariant: in the monoclinic example the flip at ab moves p_{ac} into the slot da. Kurlin (2022) normalises the square roots of the conorms, by a rule that depends on the Voronoi type, to the root invariant, a complete isometry invariant of the lattice. The invariant identifies a lattice; the closure records how its reduced cells are related.

2.4. Terms

Table 2 collects the terms used in the rest of the paper, with the section in which each is defined. Strains and prices are given in per cent throughout; a band of 7.4% is 0.074 in the formulas. A strain measures a change of metric and a price a distance between two metrics (Section 3.4). We write η for a strain and ϵ for a price band; a price band ϵ needs the reference widened at the strain $\eta = e^{\epsilon} - 1$ (Lemma A7), 7.7% for the default band of 7.4%.

Table 2
Terms and symbols.

term	meaning	defined
superbase, conorm	four lattice vectors summing to zero; the six negated pair products p_{ij}	2.1
obtuse, reduced cell	all six conorms ≥ 0; an obtuse superbase, in some labelling	2.1
zero pattern, face	the slots whose conorms vanish; the conorm vectors $p \geq 0$ with those zeros	2.1
frame	the basis in which a metric tensor or superbase is written	2.1
partial sum, parity class	the sum of some of the four vectors; a class of lattice vectors modulo twice the lattice	2.1
Niggli type I, II	a Niggli cell with three acute angles, or with none	3.1
flip, swap	the operators of Table 1; the 48 relabellings of a superbase	2.2
sort	one of Delone's 24 classes of lattices by zeros and equalities among the conorms	2.3
labelling	a choice of which three vectors of a superbase, in which order and common sign, are a, b, c	3.1
closure $\mathcal{C}(\mathbf{G})$	the obtuse superbases of a lattice in all their labellings, with the matrices between them	3.1
groupoid; object, arrow, loop	a set of objects with invertible arrows between them that compose; here metric tensors of labelled reduced cells, unimodular matrices between them, and the arrows from an object to itself	3.1
action groupoid	all metric tensors as objects and all unimodular matrices as arrows; the closure is its part on one lattice's reduced cells	3.1
visible symmetry, shape	a symmetry that relabels a reduced cell; the superbases whose conorms agree up to a swap	3.1
$\mathcal{U}$	the 6096 integer matrices between two reduced cells of some lattice	3.3
strain η, $\|\cdot\|_{\mathbf{G}}$	$\|\mathbf{G}^{-1/2}\boldsymbol{\Delta}\mathbf{G}\,\mathbf{G}^{-1/2}\|_F$, equation (2)	3.4
geodesic distance D	the affine-invariant distance between two metrics, equation (3)	3.4
depth	the strain from a metric to the nearest metric that makes a superbase obtuse	
price, displacement	the geodesic distance, after the best common scale, between a metric and the one an operator predicts; for a self-map, its displacement	3.4,
band ϵ	the price within which an operator is accepted; the reference is widened at $\eta = e^{\epsilon} - 1$	3.4
reference, new cell, candidate	the stored cell, the cell to be reindexed against it, and an operator between them within the band	4.2
widened closure $\mathcal{S}_\eta$	the superbases of depth at most η	
near-symmetry, H_ϵ	a unimodular self-map of small price; the proper ones of price at most ϵ	3.4
pseudo-symmetry	a symmetry of a nearby lattice of higher symmetry, a near-symmetry of the measured one	
μ_c	the squared length of the shortest vector of parity class c	
layer $\mathcal{N}_k$	$\mathcal{N}_1 = \mathcal{U}F_0, \mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$	3.4
adjacency radius r_k	the strain at which a superbase outside $\mathcal{N}_k$ first becomes obtuse; searched up to 25%	3.4
difference-wall member	a member of the widened set reachable from the reduced cell only through a non-member	4.1
indexing class	a class of the Laue group (point group for anomalous data) among the candidates	3.4
obliquity δ	Le Page's angle between a lattice row and the nearest reciprocal row	
non-isomorphism	the price between two crystals of one form	4.3

3. Theory

3.1. Definition

A lattice has no single preferred reduced cell. It has a finite set of obtuse superbases, and two kinds of integer matrix relate them: symmetries, which map a setting to itself, and reindexings, which map one setting to another. A groupoid holds both (Brandt, 1927; Higgins, 1971). A *labelling* of an obtuse superbase is a choice of which three of its vectors, in which order and with which common sign, are called a, b, c; each obtuse superbase has 48 labellings, joined by swaps, and each gives a metric tensor. The *Selling closure* $\mathcal{C}(\mathbf{G})$ of a lattice has these metric tensors as objects and, as arrows from $\mathbf{G}$ to $\mathbf{G}'$, the unimodular matrices $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}'$: the full subgroupoid, on finitely many objects, of the action groupoid of $GL_3(\mathbb{Z})$ on positive definite matrices (Weinstein, 1996). Elsewhere a *superbase* is the unlabelled set of four vectors, and the closure is counted by its n superbases. The arrows from an object to itself are the lattice point group in that basis (Himes & Mighell, 1987); the arrows between two objects are one coset of it (Flack, 1987); and the arrows are known in advance: a lattice has $n = 1$, 2, 3, 6 or 16 obtuse superbases according to its zero pattern (Kurlin, 2022; Lemma A2), every arrow is a word of flips at zero conorms followed by a swap (Lemma A3), and every arrow lies in one fixed set $\mathcal{U}$ (Section 3.3). The symmetries that merely relabel a superbase are its *visible* symmetries, the superbases in its orbit are those of the same *shape*, and the order of the point group is the number of visible symmetries times the number of superbases of that shape; every other symmetry carries the reduced cell to a different reduced cell. A reduced cell here is always an obtuse superbase; a Niggli cell of type II (no acute angle) is one, and a Niggli cell of type I (three acute angles) is not.

3.2. One monoclinic cell

Take $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with conorms

$$p_{da} = 639.5, \quad p_{db} = 1600.0, \quad p_{dc} = 2239.5,$$

$$p_{ab} = 0, \quad p_{ac} = 260.5, \quad p_{bc} = 0 \text{ Å}^2.$$

The two zeros are the right angles α and γ at the twofold axis b, present in every primitive monoclinic lattice. The closure has six superbases (Table 3). Each has two zero conorms, each flip at a zero leads to a neighbour, and the six form a ring M_0–M_1–M_5–M_4–M_3–M_2–M_0 (Fig. 1). They fall into three shapes. $\{M_0, M_4\}$ is 30, 40, 50 Å, $\beta = 100°$; $\{M_1, M_3\}$ is 30, 40, 53.7 Å, $\beta = 113.4°$, the cell on a and $a + c$; $\{M_2, M_5\}$ is 50, 40, 53.7 Å, $\beta = 146.6°$, the third cell choice, on c and $a + c$. These are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16).

Table 3
The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: six obtuse superbases, each up to the order and common sign of its vectors, the flips at zero conorms that connect them, and where the four symmetries of the point group $2/m$ carry M_0.

object	four vectors	flips lead to	symmetries carry M_0 here
M_0	$-a-b-c,\ a,\ b,\ c$	M_1, M_2	1, $\bar{1}$ (swaps of M_0)
M_1	$-b-c,\ -a,\ b,\ a+c$	M_0, M_5	
M_2	$-a-c,\ a+b,\ -b,\ c$	M_0, M_3	
M_3	$a+c,\ b-c,\ -b,\ -a$	M_2, M_4	
M_4	$a-b+c,\ -c,\ b,\ -a$	M_3, M_5	2 along b, m normal to b
M_5	$a-b,\ c,\ b,\ -a-c$	M_1, M_4	

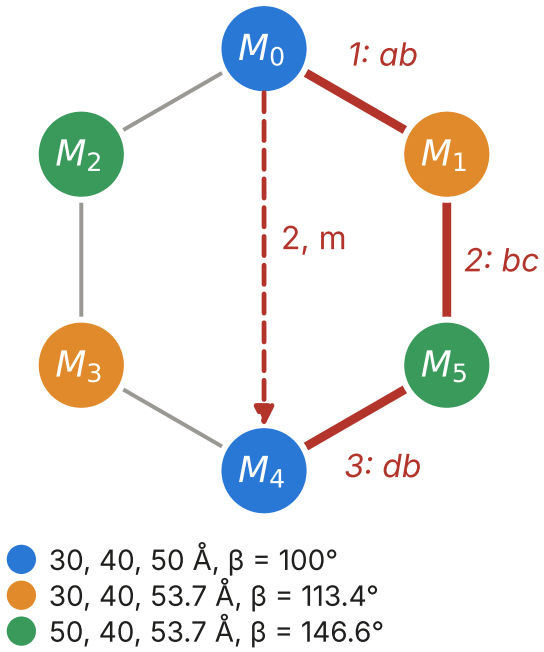


Figure 1
The closure of the 100° cell: six superbases on a ring, joined by flips at the two zero conorms and coloured by cell shape. The red path is the flip word *ab.bc.db* from M_0 to M_4, each flip read in the labelling current at that step. The twofold and the mirror carry the vectors of M_0 to those of M_4 along this path, followed by a swap; they relabel no superbase on the way, so they are invisible in the reduced cell.

The point group is $2/m$, and the last column of Table 3 shows how its four operations act on the six superbases. The identity and the inversion are swaps of M_0: they relabel it. The twofold sends $(a,b,c) \mapsto (-a,b,-c)$ and so carries the four vectors $(-a-b-c, a, b, c)$ to $(a-b+c, -a, b, -c)$, which are the vectors of M_4, three flips away. As an arrow of the closure each symmetry is a loop at the metric tensor of M_0; as a word in flips and swaps the twofold is the flips ab, bc, db — the path $M_0 \to M_1 \to M_5 \to M_4$ — followed by the swap that renames the vectors of M_4 so that its metric tensor is that of M_0, and the mirror is the same flips with that swap composed with the change of common sign. Half of the point group is therefore invisible in the reduced cell: the stabiliser of M_0 has two elements, and the orbit of M_0 is $\{M_0, M_4\}$, as the count $4 = 2 \times 2$ requires. "Visible" means only that an operation relabels the reduced superbase; a search over lattice directions, such as Le Page's, finds the twofold either way. Two measurements that reduce to M_0 and to M_1 report different cell parameters and are related by no symmetry, only by the flip at ab.

3.3. The universal operator set and the size of its entries

The arrows can be characterised with no lattice at all. An integer matrix $\mathbf{W}$ acts on the six conorms by a fixed integer 6×6 matrix, and $\mathbf{W}$ is an arrow of the closure of every lattice on a face exactly when that matrix maps the face into the cone of non-negative conorms, a sign condition on its entries (supporting information, Section S3). The union over the 38 faces is the set $\mathcal{U}$. It has $6096 = 48 \times 127$ elements with entries in $\{-3, \ldots, 3\}$ and is closed under inversion. By Lemma A2 and the exhaustive enumeration of its window, it contains every arrow between two reduced cells of any lattice; this, and the converse that every element of $\mathcal{U}$ is an arrow for every lattice on some face, are machine-checked (Section 3.5). The cone of non-negative conorms is Voronoi's domain of the principal perfect form (Voronoi, 1908; Schürmann, 2009), and $\mathcal{U}$ is the set of unimodular matrices that carry it to a domain meeting it within the positive definite metrics; the 127 superbases are those domains, and we have not found their number stated. For a reduced superbase F_0 of a lattice, the superbases $\mathcal{U}F_0$, 127 up to relabelling, are all the superbases that can be obtuse together with F_0 for some metric. This set carries the completeness statements of Section 3.4, and its entries say how large a search over integer matrices has to be (Section 5.1).

The arrows of a closure depend only on the zero pattern of the reduced cell. A superbase is obtuse exactly when each of its partial sums is shortest in its parity class. By Lemma A1, written relative to F_0, that holds exactly when certain integers vanish on the slots where F_0's conorms are non-zero. A superbase obtuse for one metric in the relative interior of a face is therefore obtuse for every metric there (machine-checked), one generic metric per face describes every lattice of that face, and the 38 faces cover all lattices. The size of the entries depends on a convention, since relabelling either cell changes them: every element of $\mathcal{U}$ is a swap, times a matrix with entries in $\{-1, 0, 1\}$, times a swap. By enumeration (Lemma A2), with the convention of Table 4: With no zero conorm, one zero, or two zeros on disjoint pairs, every operator has entries in $\{-1, 0, 1\}$. With two zeros sharing a vector or with three zeros, the entries reach 2 when a zero sits on a slot of $d = -a-b-c$, and 3 for a path of three zeros with two on d, such as conorms $(0, 0, 3, 2, 0, 5)$ with the obtuse basis $(-3a-2b-c,\ a,\ a+b)$. A relabelling of the reduced cell that moves its zeros off d brings every operator into $\{-1, 0, 1\}$ wherever such a labelling exists: for one zero, two zeros sharing a vector and a triangle of three (two zeros on disjoint pairs need only entries of 1 in any case). A path of three zeros touches all four vectors, so no labelling keeps it off d, and in the best of the 48 labellings an operator still needs an entry of 2 (`face_labellings.py`). Entries of 3 are there-

fore removed by relabelling the first cell; entries of 2 on a path are not removed by relabelling the first cell alone; relabelling both cells removes them.

Table 4
The largest entry of an operator from a reduced cell to another reduced cell of the same lattice, by kind of face and by whether a zero conorm sits on a slot of d, over all 38 faces: the first cell is taken as reduction returns it, in its own basis, and the second in its worst labelling. By enumeration (Lemma A2).

zero conorms	zeros avoid d	a zero on d
none	1	
one	1	1
two on disjoint pairs (O3, cF, R3, M3, A3)	—	1
two sharing a vector (mP, oS, hP)	1	2
three, a triangle (oP, tP, cP from an orthogonal cell)	1	2
three, a path (the same lattices from a face-diagonal cell)	—	2 or 3

3.4. Measured cells

A measured cell is never exactly on a zero conorm: a right angle is measured as 90.25°, and the conorm is a small number of either sign. The flips that joined the superbases of Table 3 now lead to superbases with a slightly negative conorm, Selling reduction rejects them, and the exact closure of the measured metric has one superbase instead of six. The closure has to be widened.

Two measures of mismatch. Two measures are used, both the same in every basis. The *strain* of a perturbation $\Delta\mathbf{G}$ of a metric $\mathbf{G}$ is

$$\|\Delta\mathbf{G}\|_{\mathbf{G}} = \|\mathbf{G}^{-1/2}\,\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F. \tag{2}$$

It is unchanged by a change of basis, since $\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$ and $\Delta\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\Delta\mathbf{G}\mathbf{P}$ leave it as it is, and by a common scale of the cell. A perturbation of strain η changes the dot product of any two lattice vectors by at most $\eta|u||v|$ (Proposition A4). The strain measures how far a metric must move to make a superbase obtuse, a convex problem in its six conorms. The *geodesic distance* between two metrics is

$$D(\mathbf{G}_1, \mathbf{G}_2) = \Big(\sum_i \ln^2 \boldsymbol{\lambda}_i\Big)^{1/2}, \tag{3}$$

with λ_i the eigenvalues of $\mathbf{G}_1^{-1/2}\mathbf{G}_2\mathbf{G}_1^{-1/2}$, the affine-invariant distance on positive definite matrices (Pennec *et al.*, 2006). It agrees with the strain to first order, it is a true distance (symmetric, with the triangle inequality), and every change of basis is an isometry for it. It measures how far apart two cells are, and it prices operators.

Table 5 converts the classical tolerances. For Le Page's twofold criterion (Le Page, 1982) the conversion is exact: a twofold whose obliquity, the angle between a lattice row and the nearest reciprocal row, is δ sits at a geodesic distance $\sqrt{2}\,\operatorname{artanh}(\sin\delta)$ from the metrics it would leave invariant, and its price is twice that (Proposition A9). One degree of obliquity is a price of 4.9%, and Le Page's 3° a price of 14.8% (Table 5), for every twofold separately.

Table 5
Classical tolerances as strains and prices. The first two rows give, to first order, the smallest strain of such a change, reached for an orthogonal pair (first row) or an axis normal to the other two (second row); the last two rows are exact for a twofold along a lattice row, in every cell (Proposition A9).

tolerance	strain or price	value
angle between two vectors changed by δ, edges fixed	at least $\sqrt{2}\sin\delta$	7.4% at 3°
edge changed by a fraction f, angles fixed	at least $2f$	10% at 5%
twofold of obliquity δ: geodesic distance to the metrics it fixes	$\sqrt{2}\,\operatorname{artanh}(\sin\delta)$	7.4% at 3°
twofold of obliquity δ: its price	$2\sqrt{2}\,\operatorname{artanh}(\sin\delta)$, 4.9% per degree	14.8% at 3°

The widened set. The *widened closure* of a metric $\mathbf{G}$ at strain η is

$$\mathcal{S}_\eta(\mathbf{G}) = \big\{\, F \;:\; F \text{ is obtuse for some metric within strain } \eta \text{ of } \mathbf{G} \,\big\}. \tag{4}$$

Its superbases are the reduced cells of every lattice consistent with the measurement at that tolerance. Each superbase has a *depth*, the strain from $\mathbf{G}$ to the nearest metric that makes it obtuse, and belongs to $\mathcal{S}_\eta(\mathbf{G})$ when its depth is at most η. Equation (4) names no starting cell, order of operations or frame, so $\mathcal{S}_\eta(\mathbf{G})$ is the same whoever computes it. Each operator $\mathbf{P}$ from a metric $\mathbf{G}_1$ to a metric $\mathbf{G}_2$ has a *price*, the geodesic distance between $\mathbf{G}_2$ and $s\mathbf{P}^{\mathsf{T}}\mathbf{G}_1\mathbf{P}$ minimised over a common scale s, which removes the trace of the logarithm in equation (3). The price is zero for every arrow of an exact closure, and for a unimodular self-map the scale changes nothing.

It is finite. The seven partial sums of an obtuse superbase are shortest vectors of their parity classes (Section 2.1). With μ_c the squared length of the shortest vector of class c, every vector of a superbase in $\mathcal{S}_\eta(\mathbf{G})$ lies in a ball of squared radius $\frac{1+\eta}{1-\eta}\mu_c$ (Lemma A6), so $\mathcal{S}_\eta(\mathbf{G})$ is enumerated exactly, with no window of coordinates to assume.

Widening one side suffices. If an operator $\mathbf{P}$ from a reference to a new cell has price at most ϵ, the new cell's reduced superbase, carried back by $\mathbf{P}$, lies in the reference's widened closure at strain $\eta = e^\epsilon - 1$ (Lemma A7). So only the reference is widened, and the new cell is reduced once.

Layers and the lookup. Call two superbases adjacent when some metric makes both obtuse. The superbases adjacent to F_0 are the first layer $\mathcal{N}_1 = \mathcal{U}F_0$, and $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$. When the widened set at the strain $e^\epsilon - 1$ that a price band ϵ needs lies in $\mathcal{N}_1$, the matrices of $\mathcal{U}$ between the two Selling-reduced cells already contain every operator within ϵ (Remark A8); the enumeration reports the smallest layer that holds the set, which certifies this. The *adjacency radius* $r_k(\mathbf{G})$, the strain to the nearest metric for which a superbase outside $\mathcal{N}_k$ is obtuse, searched up to 25%, certifies at once every price band ϵ with $e^\epsilon - 1 < r_k$. Nei-

ther certificate makes the enumeration unnecessary; the lookup is a fixed table, deposited with this paper, that turns each new cell into one batch of matrix products once a reference is known to be in the first layer.

Exact closures bound the widened set. By equation (4), $\mathcal{S}_\eta(\mathbf{G})$ is the union of the closures $\mathcal{C}(\mathbf{G}')$ of the lattices $\mathbf{G}'$ within strain η of $\mathbf{G}$: its objects are those of exact closures, but its matrices need not compose within the tolerance, so it is not itself a groupoid. The price of a self-map h of a metric is its *displacement* $D(\mathbf{G}, h^{\mathsf{T}}\mathbf{G}h)$ under the isometric action of $GL_3(\mathbb{Z})$: the displacement function of the action (Bridson & Haefliger, 1999, Chapter II.6). The self-maps, proper and improper, of price at most ϵ form a finite, symmetric set that contains the point group, and two of them compose to one of price at most the sum. The closure of the nearest symmetric lattice bounds this set from both sides: if $\mathbf{G}_0$ is at geodesic distance ν from $\mathbf{G}$, the point group of $\mathbf{G}_0$ consists of self-maps of $\mathbf{G}$ of price at most 2ν, and every self-map of $\mathbf{G}$ of price at most ϵ is a self-map of $\mathbf{G}_0$ of price at most $\epsilon + 2\nu$ (Proposition A10(v)). The groupoid computes the matrices, exactly and once, as $\mathcal{U}$; the tolerance bounds which of them a measurement can reach.

Cosets and classes. Let H_ϵ be the proper unimodular self-maps of the reference of price at most ϵ, and p_0 the price of the true operator $\mathbf{P}_0$. Because the price is a distance, every candidate is $h\mathbf{P}_0$ with $h \in H_{\epsilon+p_0}$, and every $h \in H_{\epsilon-p_0}$ gives a candidate; if no self-map has a price in $(\epsilon, 2\epsilon]$, H_ϵ is a group, and if none has a price in $(\epsilon - p_0, \epsilon + p_0]$, the candidates are exactly the coset $H_\epsilon\mathbf{P}_0$ (Proposition A10). Two candidates that differ on the left by an element of the reference's Laue group $\mathcal{L}$ index the crystal the same way, so the distinct indexings are the classes of the proper part $\mathcal{L}^+$ of $\mathcal{L}$ among the candidates. For anomalous data from an acentric crystal the point group replaces the Laue group. The guarantee is that the true operator is among the candidates; the price orders the classes but does not pick the true member, which is a matter for the intensities.

The price of pseudo-symmetry. A symmetry g of a nearby lattice enters H_ϵ when its price is at most ϵ. For a twofold the price is exactly twice the geodesic distance from the measured metric to the metrics V_g that g leaves invariant; for a rotation of order n the price is at most twice that distance, and to first order near V_g the ratio is $\sqrt{3}$ for a threefold, between $\sqrt{2}$ and 2 for a fourfold and between 1 and $\sqrt{3}$ for a sixfold (Proposition A9). For twofolds one threshold on the price therefore serves every sort.

3.5. What is proved

Table 6 separates what is proved from what rests on an exhaustive computation and what is checked. "Machine-checked" marks statements proved in Lean 4 (version 4.35.0) with Mathlib (de Moura & Ullrich, 2021; mathlib Community, 2020; Mathlib release v4.35.0). The finite steps are evaluated with `native_decide`, which trusts the Lean compiler as well as the kernel; the axioms of each theorem are listed in the supporting information (Section S7). The Lean copy of $\mathcal{U}$ is tied to the deposited table by its SHA-256 hash, and the enumeration that completes Lemma A2 was also run independently in Python. The affine-invariant distance enters the formal statements only through the properties stated in Section 3.4 and, for Proposition A9(i), the existence and uniqueness of midpoints. The proofs rest on Selling's identity, the characterisation of obtuse superbases by Voronoi vectors (Conway & Sloane, 1992; at a zero conorm a partial sum is a Voronoi vector that is not strict) and Selling reduction. Exact closures and the closures of the 24 sorts were checked against brute force over bases with entries in $[-2, 2]$, a window that contains a labelling of every member of every exact closure (by the enumeration that completes Lemma A2), and widened sets against the ball enumeration of Lemma A6.

Table 6
What is proved, what rests on an exhaustive computation, and what is checked or refuted.

statement	status
every arrow is flips at zero conorms and a swap	proved (Lemma A3)
the members of each closure lie in a finite window	proved (Lemma A2); machine-checked, in the basis (a, b, c) with coordinates in $[-7, 7]$
$\mathcal{U}$ contains every arrow between reduced cells of any lattice; every member of $\mathcal{U}$ is an arrow on some face; arrows depend only on the zero pattern; the entry sizes by zero pattern	computer-assisted: the window enumerated over all 38 faces; the first three machine-checked, with Lemma A1, against the deposited table
the 24 sorts	a definition and an enumeration, matched to Delone's patterns by computation (Definition A5)
the strain bounds every dot product	proved (Proposition A4)
the widened set is finite and its candidates enumerated exactly	proved (Lemma A6); membership by a floating-point depth, with a relative margin of 10^{-9}
in the first layer, the lookup of $\mathcal{U}$ is complete	proved (Remark A8, from Lemma A7, the completeness of $\mathcal{U}$ and its closure under swaps), the last two machine-checked; the radii of Table 8 certified in exact arithmetic for the stored conorms, those of the archive statistics computed in floating point, up to 25%
widening the reference alone suffices, at $\eta = e^{\epsilon} - 1$	proved (Lemma A7)
a twofold's price is twice its distance to the metrics it fixes, and $2\sqrt{2}\,\operatorname{artanh}(\sin\delta)$	proved, exactly (Proposition A9); the bound of part (i) machine-checked for any isometric action, and the fixed midpoint of a twofold given that midpoints exist and are unique
the ratios of price to distance for orders 3, 4, 6	proved in the strain at a symmetric metric; first order for the price (Proposition A9)
a type II Niggli cell is obtuse, with its small conorms off d below a stated size, near two or three right angles; a type I metric near a three-zero lattice reduces to a path through d from any basis	proved (Proposition A11), parts (i) and (ii) machine-checked; type I near two right angles, and reduction from other bases, observed
candidates lie between two cosets of self-maps; H_ϵ is a group in a gap of the prices; the near-symmetries lie between the point group of the nearest symmetric lattice and its self-maps within $\epsilon + 2\nu$	proved, exactly (Proposition A10); machine-checked for any isometric action
the layers of 127 and 2755 superbases; the largest entries after relabelling; Lemma S4.4 and so Theorem S4.5 of an absolute band on the conorms (supporting information, Section S4)	computer-assisted enumeration, the layers also recomputed independently

4. Methods

All methods are implemented in the module `agentsg.cell.strain_closure` of `agentsg` (Zwart, 2026), version 0.5.0; the test cases and the PDB analysis are scripts in the same repository (supporting information, Section S2).

4.1. Computing the widened set

For a measured cell and a strain $\eta < \frac{1}{4}$: Selling-reduce the cell to a superbase F_0; enumerate the lattice vectors in the seven balls of Lemma A6 and form the superbases they make; keep those of depth at most η, with each one's depth and the operator $\mathbf{W}$ from F_0 to it. The candidates are exact for every cell; the depth that decides membership is computed in floating point, with a relative margin of 10^{-9}, so only a member within that margin of the boundary could be misclassified. Walking from the reduced cell, flipping wherever a conorm is within a band of zero, is not: a member can be reachable only through a non-member, when two small conorms lie at, say, 1.2 and 0.5 times the band and only the second of two flips returns into it. We call such a member a *difference-wall member*, since it lies across the wall where the two small conorms are equal; walks that miss such members are compared in Table S3 of the supporting information.

Ties take no band. Every sort lies on one zero pattern, and every symmetry of every lattice on that pattern is one of its arrows (Lemma A3), so a near-equality — the $a \approx b$ of a pseudo-tetragonal cell — never adds a superbase: the operator that would be its symmetry is already enumerated, and its price decides whether it is accepted.

4.2. Reindexing

A reference R is stored as its widened closure, the operators $\mathbf{W}_k$ from its reduced cell to the members. A new cell N is reduced once, to a basis B_N in its own coordinates; B_R is the reduction matrix of the reference. For each stored $\mathbf{W}_k$ and each of the 48 swaps $\mathbf{Q}$ the candidate operator from the reference's input basis to the new cell's is

$$\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q}\, B_N^{-1}, \tag{5}$$

so that $(a_N\, b_N\, c_N) = (a_R\, b_R\, c_R)\mathbf{P}$, $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$, and Miller indices transform as $h_N = \mathbf{P}^{\mathsf{T}} h_R$: for the flip at ab of Table 1, $\mathbf{P} = \mathbf{F}_{ab}$, the new cell is $(-a,\ b,\ a + c)$, and the reflection $(1, 0, 0)$ of the reference is $(-1, 0, 1)$ in the new cell. Candidates of price at most the band ϵ are returned; the reference is widened at $e^{\epsilon} - 1$, so the true operator is among them whenever its price is at most ϵ (Lemma A7). Three conventions complete the procedure. Only operators of positive determinant are returned: cell parameters carry no hand, so the convention is that both bases are right-handed, and an inverted geometry must be caught before the cell reaches this step. A fitted scale that changes the edge lengths by more than 10% is rejected, and so are two primitive cells whose volumes differ by more than that allows; a wrong centring symbol or a doubled axis lands here. The limit is far looser than any shape band, so isotropic non-isomorphism is never penalised. An obverse–reverse relation between two rhombohedral lattices keeps the volume and

the hexagonal cell, so no cell-based test sees it; it is a matter for the intensities. A centred reference is first taken to a primitive cell (a rhombohedral cell on rhombohedral axes is already primitive) and the centring matrix composed with the operator found, which is then rational in the conventional basis. When the reference's widened set lies in $\mathcal{N}_1$ the candidates can come from $\mathcal{U}$ directly (Table 7, step 3); $\mathbf{W}$ is then a column operator (only 768 of the 6096 transposes lie in $\mathcal{U}$), and since Selling reduction may return a left-handed basis, a proper $\mathbf{P}$ needs the 3048 matrices with $\det \mathbf{W} = \det B_R \det B_N$.

Table 7
Reindexing a new cell against a reference within a price band ϵ.

step	action
1	Choose the price band ϵ (Section 4.3: 7.4% for completeness against non-isomorphism; 9.9% or 14.8% to list the alternatives that a 2° or 3° lattice-symmetry search would test). Reference: take the primitive cell, Selling-reduce it (B_R, F_0; a flip fires only at a conorm below -10^{-9} times the largest squared length, so exact zeros do not oscillate), and enumerate its widened set at strain $\eta = e^{\epsilon} - 1$ from the seven balls of Lemma A6; store the members, the operators $\mathbf{W}_k$ to them, and the smallest layer holding the set. A larger band needs the reference widened again.
2	New cell: take its primitive cell, with its own centring symbol (a rhombohedral cell on hexagonal axes reduced through its rhombohedral primitive cell, as for the reference), and Selling-reduce it once (B_N). Refuse if the primitive volumes differ by more than the scale limit allows, between 0.9^3 and 1.1^3 for a 10% limit on the edges.
3	Candidates: $\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q} B_N^{-1}$ for every member and swap $\mathbf{Q}$ (equation 5); if the set lies in $\mathcal{N}_1$, equivalently $\mathbf{P} = B_R \mathbf{W} B_N^{-1}$ for the 3048 $\mathbf{W} \in \mathcal{U}$ (column operators) with $\det \mathbf{W} = \det B_R \det B_N$.
4	Keep $\det \mathbf{P} > 0$, price each by the geodesic distance after the best common scale, and keep those within ϵ whose scale changes the edges by at most 10%.
5	Group the candidates by the left action of the proper Laue group (point group for anomalous data); if the space group is itself in question, group by the proper holohedry of the reference's Bravais lattice, or report the candidates ungrouped. Report each class with its cheapest operator and price, read the prices against the non-isomorphism of the crystal's Bravais type (Table 10), and flag the result when the cheapest class costs more than its 95th percentile.
6	When more than one class lies within the band, decide with the intensities (Brehm & Diederichs, 2014; Gildea & Winter, 2018).

4.3. Choosing the band

The band is a tolerance on how much two measurements of one crystal may differ. For macromolecular crystals that difference is dominated by non-isomorphism, not by the measurement error that the standard uncertainties of a cell describe. We measure non-isomorphism in the PDB (Section 4.4) and take the scale of the band from its upper percentiles, which depend on how a crystal form is delimited and on the Bravais type (Section 5.4). A candidate's price is read against that distribution: a class priced at the median non-isomorphism is as close to the reference as a typical second crystal of the same form, and the metric cannot exclude it. The tables of this paper use a price band of 7.4%, the implementation's default, the strain of one cell angle changed by 3° (Table 5). As a price band it admits a pseudo-twofold only up to an obliquity of about 1.5°, so it is narrower than the lists of alternatives that data-processing programs test with the intensities: Pointless and `dials.symmetry` search lattice symmetry within 2° by default and the twin analysis of `phenix.xtriage` within 3°, prices of 9.9 and 14.8%. The band defines completeness against non-isomorphism; to enumerate those alternatives the band is set to 9.9 or 14.8%, and the reference widened at 10.4 or 16.0% strain. Any band is a percentile of non-isomorphism, not a bound on it: the pairs whose cells differ by more than the band, about 4% of pairs of one crystal form at 7.4%, are not reindexed within it (Section 5.4). For small-molecule data, whose cells are measured precisely and rarely non-isomorphous, a band of a few times the strain implied by the esds is the natural choice (`strain_from_esds` in the implementation); it is not tested here.

4.4. Tests

Constructed cases. The guarantee is tested on cases built where the closure predicts trouble (supporting information, Section S2). The benchmark has 780 primitive cells with a random metric error (median strain 0.1%) and a random proper reindexing: 75 measured in another shape of the same lattice, 660 near a specialisation, 20 whose only reduced superbase is a difference-wall member, 16 needles and plates and 9 of the boundary sorts. The stress family puts independent errors on both cells of 60 references, scaled so that the true operator's price is 0.5 and 0.9 times the band. A further family has 300 lattices with two right angles sharing a vector or three right angles, each measured twice with an error of 0.1 to 2%, the reference given as its Niggli cell or in a random basis, and 60 lattices measured in a member that needs an entry of 3. In every case the true operator is known, and the test is whether it is among the candidates within the band. The same cases were also given to re-implementations of the searches that data-processing programs use (Section 5.2; supporting information, Section S8, Table S13).

The PDB. The snapshot of 206 214 deposited cells (Zenodo record 22986222) is used twice. Non-isomorphism is measured between entries of one crystal form, taken as entries with the same space group whose identifiers are adjacent and share their first three characters, so that most pairs come from one deposition series; both cells are in the standard setting of the space group, so the identity is taken as the operator, and pairs above 20% are counted as different forms. The choices behind this proxy are varied as controls: the cut, identical cells, the weight of large deposition blocks, the Bravais type, and the cheapest operator in place of the identity. Pseudo-symmetry is measured by running the cctbx lattice-symmetry analysis at 3° on the primitive Niggli cell of every entry and pricing every proper operator of the highest lattice group that is neither a symme-

try of the reported space group nor of the reported lattice's own holohedry. A null model shuffles the cell parameters within a space group. Deposited cells are refined in the reported space group, so they satisfy its symmetry exactly; what the analysis measures is how far real crystals lie from a higher lattice.

5. Results

5.1. Where reduction reaches large entries

When does reduction put a zero on d? The entries are read in the basis of the first cell, so the question is about the reference's reduced superbase, and near a lattice with three right angles the Niggli type answers it (Proposition A11). A type I measurement, whose Niggli cell has three acute angles, has no obtuse orthogonal cell, so near enough to the right angles every reduction ends on a face-diagonal cell whose zeros form a path through d, whatever basis it starts from: 402 of 402 reductions of measured near-oP cells from random bases, and 67 of 67 from the Niggli cell (`niggli_presentation.py`). A type II Niggli cell is itself obtuse with its small conorms off d, so reduction started from it stays there (83 of 83), but reduction started from another basis put a near-zero conorm on d in 370 of 498 cases. Near a lattice with two right angles a type II Niggli cell kept its right angles on ab, ac, bc in 519 of 519 measured near-mP cells, as Proposition A11(i) requires; from a type I Niggli cell the outcome depends on the order of flips: reduction put at least one near-zero conorm on d in 614 of 681, and both near-zero conorms in 316. Which type a measurement gets is decided by the signs of its small errors; in that sample 57% of near-mP measurements were type I.

5.2. Completeness on measured cells

The worked cell. The measured cell 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80° is the 100° cell of Section 3.2 with β moved to 92°, perturbed by a metric error of strain 0.82%. Its widened closure grows with the strain: one member at 0.1%, two at 0.3%, three at 0.6%, the six of the ring from 1%, and sixteen at 7.4% (Fig. 2; the members, as flip words with their conorms and depths, are in Table S6). The six members of the ring enter below 0.8%; the other ten members of the closure of the nearby orthorhombic lattice enter between 3.8 and 5.1%.

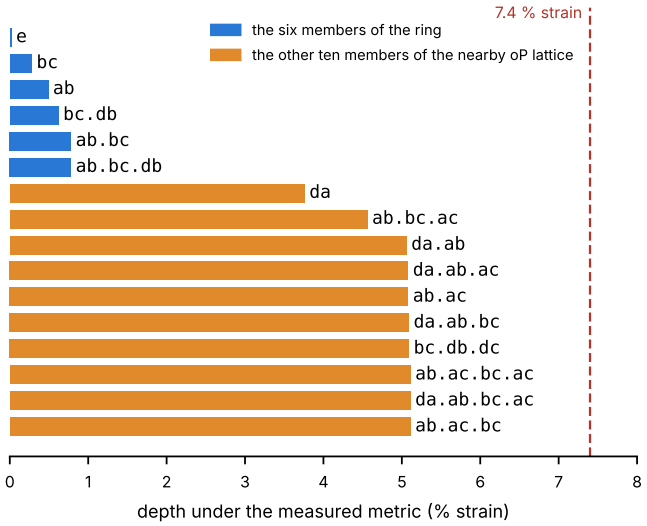


Figure 2
The sixteen members of the widened closure of the measured worked cell by depth: the six of the ring enter below 0.8% strain, the other ten members of the nearby orthorhombic lattice's closure between 3.8 and 5.1%.

Needles and plates. The exact enumeration and brute force agree on a needle 5.58 × 5.21 × 131.97 Å, $\gamma = 91.12°$, at strains from 0.05 to 5.2%, on two plates at 5.2%, where the widened sets have 20 and 28 members, and on 240 random needles, plates, isotropic and monoclinic cells. For very thin cells at large bands the ball of Lemma A6 grows quickly, and the implementation refuses balls above 5000 vectors with a message.

Adjacency radii. Table 8 gives the adjacency radii. For the worked cell and a 10 × 10 × 30 Å needle no superbase outside the first layer comes within 25%, so the lookup of $\mathcal{U}$ is complete at every band in use. For needles and plates r_1 falls with the ratio of the shortest to the longest edge, to 4–6% for the thinnest cells, and the second layer roughly doubles it; there the lookup is complete only for bands below r_1, and the exact enumeration is needed beyond. For the BLDConograph sample cells and a lysozyme cell (supporting information, Table S5 and Section S6) r_1 reaches the cap, and so it does for 99.4% of the 697 benchmark references and 98.6% of 2000 PDB cells drawn at random; every one of those 2000 has r_1 above the 7.7% that certifies the default band. The archive has a thin tail: 3HR2 and 3HQV (40 × 27 × 678 Å, $r_1 = 6.0\%$) and 6BZP (13.8 × 4.9 × 101.9 Å, 6.8%) lie below 7.7%, where the enumeration uses the second layer, and a placeholder cell of 1 × 1 × 70.8 Å (1CGM) is refused by the cap on the ball. Widening a reference and reindexing a new cell take a few hundredths of a second for ordinary cells, and about a minute for a 300 × 300 × 3 Å plate (timings in the supporting information, Section S7).

Table 8
Adjacency radii in strain (%): r_1, the smallest strain at which a superbase outside the first layer becomes obtuse, and r_2 for the second layer, searched up to 25%. Each value is certified in exact rational arithmetic, to the digits shown, for the reduced conorms of the cell as stored in double precision (`certified_radii.json`), by a dual bound for every superbase outside the layer within the cap and a feasible point for the nearest one (`certify_radii.py`).

cell	r_1	r_2
worked cell, 30.02 × 39.97 × 50.03 Å, 90.25, 92.05, 89.80°	> 25	> 25
needle 10 × 10 × 30 Å	> 25	> 25
needle 10 × 10 × 100 Å	13.9	> 25
needle 5.58 × 5.21 × 131.97 Å, $\gamma = 91.12°$	5.6	11.1
plate 100 × 100.1 × 5 Å, 90.03, 89.98, 90.05°	5.0	14.1
plate 150 × 120 × 4.8 Å, $\beta = 90.4°$	4.3	10.0

The guarantee on the test cases. The closure found the true operator in every case of every family, as Lemma A7 guarantees: 780 of 780 on the benchmark, 120 of 120 in the stress cases, 600 of 600 near two- and three-zero faces and 60 of 60 in members that need an entry of 3. The lookup of $\mathcal{U}$ found it in every case too, including 12 benchmark cases on three needle and plate references whose r_1 lies below the 7.7% that would certify it; that is not assured: against the $100 \times 100.1 \times 5$ Å plate of Table 8 ($r_1 = 5\%$), a new cell whose true operator costs 6.1% is found by the enumeration and missed by the lookup. *Existing searches.* Re-implementations of the searches in use were run on the same cases (supporting information, Table S13). The $\{-1, 0, 1\}$ window between the two Niggli cells and the cctbx similarity search found the true operator in every case of every family; the lattice-symmetry pipeline missed 1 of 780 benchmark cases, 10 of 120 stress cases and 8 of 600 near two- and three-zero faces. A $\{-1, 0, 1\}$ window between the two Selling-reduced cells, which no program uses, misses operators where Table 4 says it can: it found 24 of the 60 entry-3 cases and 219 of 300 references given in a random basis.

Why the Niggli window works. The true operator between the two Niggli cells was computed directly, with no search, for 100 lattices on each of the 38 faces and on each of the 24 sorts, measured twice with independent errors of total strain 0.5 to 7.7% and the new cell in a random basis: its entries were in $\{-1, 0, 1\}$ in all 24 800 cases, and in 3 040 further cases on strongly anisotropic lattices, the square root of the metric's eigenvalue ratio up to 98 (`niggli_window.py`, `results_niggli_window_thin.txt`). Between the two Selling-reduced cells the same operators needed an entry of 2 or 3 in 3 765 cases, all on lattices with two or three right angles, and none once composed with the lattice's point group. A Niggli cell is built on short vectors and has no fourth vector d, so it does not meet the zeros on d that make the Selling entries large. This is observed, not proved; it explains why the searches in use succeed, and the guarantee of this paper does not depend on it.

5.3. Indexing classes

Table 9 gives the classes for six references against the same crystal measured again with a 0.1% error in the length scale, which the scale fit absorbs, and a relabelling. For the exactly symmetric references the counts are the classical indexing ambiguities of merohedral crystals (Yeates, 1997; Dauter, 2003; Evans, 2006, 2011): one class for $P4_32_12$, four for $P3$, two for $P3_121$ and $P2_13$. The price adds the pseudo-merohedral classes: a $40.0 \times 40.3 \times 80$ Å $P2_12_12_1$ cell and the worked cell in $P1$ each have a second class, at 2.11% and 1.55%, twice, to first order, the cells' distances of 1.06% and 0.78% to the sorts the twofolds would make exact, from the twofold that would make the cell tetragonal or monoclinic. With proper operators only, anomalous data split no class of a chiral crystal; for an acentric crystal that is not chiral the point group replaces the Laue group, and $P4mm$ goes from one class to two and $P31m$ from two to four.

Table 9
Distinct indexings of six references at a band of 7.4% (the reference widened at 7.7%): the order of the band's group of proper self-maps H_ϵ and of the proper rotations $\mathcal{L}^+$ of the Laue group, the number of classes among the candidates, and the price of each class. The new cell is the reference with a 0.1% scale error, which the scale fit absorbs, relabelled.

reference	space group	$\lvert H_\epsilon \rvert$	$\lvert \mathcal{L}^+ \rvert$	classes	prices (%)
79.1 × 79.1 × 37.9 Å	$P4_32_12$	8	8	1	0
$a = b = 50$, $c = 80$ Å	$P3$	12	3	4	0, 0, 0, 0
$a = b = 50$, $c = 80$ Å	$P3_121$	12	6	2	0, 0
$a = 60$ Å	$P2_13$	24	12	2	0, 0
40.0 × 40.3 × 80 Å	$P2_12_12_1$	8	4	2	0, 2.11
worked cell	$P1$	2	1	2	0, 1.55

What the metric does not decide. The cheapest candidate lies in the true operator's coset of the exact lattice symmetries in 742 of the 780 benchmark cases, and is the true operator itself in 741. The 38 that fail are the 20 difference-wall cases and 18 of the 660 cases near a specialisation; in each the cheapest candidate is the true operator composed with a pseudo-symmetry, of the nearby orthorhombic corner for the difference-wall cases, that costs less than the true operator at that error. In the stress cases the cheapest candidate is in the true coset in 68 of 120, and is the true operator itself in 40. When a pseudo-symmetry costs less than the error, the price orders the candidates but cannot pick the true one; how often that happens in real crystals is the subject of the next section.

5.4. Application: the price of pseudo-symmetry in the PDB

Non-isomorphism. Between deposited cells of one crystal form (45 222 pairs) the price after the scale fit has a median of 0.98%, a 90th percentile of 4.2% and a 95th of 6.7% (95% intervals with deposition blocks resampled, 4.1–4.3 and 6.5–7.0%); without the scale fit the median is 1.6%. The upper percentiles depend on how a crystal form is delimited: with the cut between forms at 5, 10, 20 and 50% the 90th percentile is 2.5, 3.4, 4.2 and 5.6% and the 95th 3.1, 4.7, 6.7 and 11.9%. Dropping the 1 464 pairs of identical cells changes little (4.3 and 6.9%), and weighting each deposition block equally, so that fragment-screening series of up to 35 pairs count once, gives 5.0 and 8.0%. By reported Bravais type (Table 10) the 90th and 95th percentiles are 2.6 and 4.0% for tP, 2.9 and 4.6% for hP, 4.7 and 7.4% for oP, 5.1 and 8.4% for mC and 5.4 and 9.0% for mP; for the cubic types they are 0, the scale fit removing the whole difference. For aP they are 7.7 and 11.7% with the identity as the operator, and 6.1 and 9.9% with the cheapest operator, since two triclinic deposits of one crystal are often

in different reduced settings; for mP, mC and oP the cheapest operator changes them by 0.5 points or less (`results_pdb_band_controls.txt`). The scale a reindexing band has to cover for macromolecular data is therefore a few per cent: at the 20% cut, 2.6–6.1% at the 90th percentile and 4–10% at the 95th across the non-cubic types (aP with the cheapest operator); the standard uncertainties of a single cell describe its measurement error alone and do not reach it. The default band of 7.4% lies above the pooled 95th percentile and above the 90th percentile of every type except aP with the identity as the operator (7.7%; 95% interval 6.4–9.5%). The intervals of every percentile are in the supporting information (Table S14); the oF row rests on 31 pairs in 22 blocks and is not informative, and the cubic rows are zero by construction.

Table 10

Non-isomorphism in the PDB by reported Bravais type: the price (%, after the scale fit) between deposited cells of one crystal form, taken as id-adjacent entries of one space group below 20% unfitted strain, with the identity as the operator; for the low-symmetry types also the 90th and 95th percentiles with the cheapest operator. The cells are deposited cells refined with the reported symmetry imposed, so only their free parameters carry error: cells from indexing, or from serial and multi-crystal data, need wider bands. For the cubic types the scale fit removes the whole difference; the oF row rests on 22 deposition blocks. Bootstrap intervals are in Table S14.

type	pairs	median	90th	95th	90th / 95th, cheapest operator
aP	902	2.06	7.66	11.68	6.14 / 9.89
mP	7 077	1.27	5.38	9.02	5.29 / 8.80
mC	4 932	1.49	5.06	8.40	4.82 / 7.90
oP	13 615	1.11	4.70	7.43	4.65 / 7.39
oS	2 271	1.01	3.57	5.50	—
oI	1 353	1.22	5.12	7.91	—
oF	31	1.99	5.87	7.54	—
tP	5 174	0.52	2.59	4.00	—
tI	805	0.65	2.76	4.91	—
hR	1 270	0.68	3.54	5.86	—
hP	6 817	0.57	2.87	4.56	—
cP	363	0.00	0.00	0.00	—
cI	385	0.00	0.00	0.00	—
cF	211	0.00	0.00	0.00	—

Pseudo-symmetry. Le Page's analysis at 3° finds a higher lattice than the reported one for 43 201 of the 206 141 entries analysed (21%); 31 placeholder cells with an edge below 2 Å and 42 cells the cctbx analysis did not accept are left out. For 545 (0.3%) the higher lattice is exact. Operators of the reported lattice's own holohedry, taken from its Bravais type, which a merohedral space group such as $R3$ or $P4_1$ leaves out of its point group, are not counted as pseudo-symmetry. The operators are those of the cctbx analysis, priced as in this paper. For the 42 656 entries without an exact higher lattice the largest price matches $2\sqrt{2}\,\mathrm{artanh}(\sin\delta)$ for the group's largest obliquity δ (ratio 1.000 at the 10th, 50th and 90th percentiles), consistent with Proposition A9 when a twofold is the most expensive operator, and a check of the code, and the cheapest operator has a median price of 6.7%. Fig. 3 sets these prices against non-isomorphism: the cheapest alternative lies within the 90th percentile of non-isomorphism for 35% of the pseudo-symmetric entries and within the 95th for 50%. Over all entries, 7.5% (95% interval 7.4–7.7%) have an alternative, exact or priced, within 4.2% and 10.6% (10.4–11.0%) within 6.7%; with the band of the entry's own Bravais type, the pooled band standing in for a type with fewer than 200 deposition blocks, 8.6% (8.4–9.0%) and 12.5% (12.0–13.0%).

Against chance. Shuffling the free cell parameters among the entries of one space group, with and without keeping cells of one volume together, shows that pseudo-symmetry within 3° is reached by chance almost as often as in real cells, while near-exact pseudo-symmetry is not chance: the share whose cheapest alternative costs at most 1% is 10–11% for real $P1$ cells against 2–4% shuffled, and 9% against 5–6% for $P2_1$ (supporting information, Section S9). The obliquities of the priced alternatives are spread nearly evenly below 3°, so median prices of 6–7.5% follow from the 3° window, and the match of the largest price to $2\sqrt{2}\,\mathrm{artanh}(\sin\delta)$ is a check of the implementation against Proposition A9 rather than a finding.

What the cell decides. An alternative within the band is one the cell does not exclude; it is not one the cell prefers. Reindexing the second cell of 3 000 randomly chosen same-form pairs against the first, with the first cell's space group and a band of 7.4%, a wrong class lies within the band at a positive price in 7.9% of pairs, and at exactly the identity's price in 17%: the classical merohedral ambiguities within the reported lattice type, such as $P3$ or $P4$ in a lattice of higher holohedry, which no cell can resolve. A wrong class is strictly cheaper than the identity's in 36 pairs (1.2%; 0.8–1.6% with deposition blocks resampled): in 12 by at most 0.05 points, below the precision of a deposited cell; in 20 through a relabelling of axes, which mixes deposits that use different axis orders, where the identity is not the true operator, with genuine near-ties of pseudo-tetragonal cells; and in 4 otherwise (0.13%, 0.03–0.27%). Telling these apart needs the deposited models, which we did not compare. In 125 pairs (4.2%) the two cells differ by more than the default band, so the identity, the true operator for two deposits in the standard setting, costs more than 7.4% and is not returned; this is the upper tail of non-isomorphism (Table 10; 4.3% of all 45 222 same-form pairs exceed 7.4%, with a median price of 10.7%), and in 9 of these pairs (0.3%) only a wrong class is within the band. So for about one pair in thirteen the cell leaves a priced alternative open, and it ranks a wrong indexing first in between about 0.1% and 1.2% of pairs, depending on how the relabellings are read. This matters when cells are compared, in reindexing against a reference or in merging, not for a single data set.

Where. Monoclinic and triclinic entries carry most of the share: 18% of the 35 364 mP entries and 16% of the 8 491 aP entries have an alternative within 4.2%, against 7% of oP and 1% of tP entries. Among the transitions, mP to hP is nearly always undecidable from the cell (median price 0.71%, 91% within 4.2%), the configuration in which pseudo-merohedral twinning is possible; for most transitions the median price of the cheapest alternative is 6–7.5%, so the sort decides how often a pseudo-symmetry occurs, not what it costs (supporting information, Tables S8 and S9).

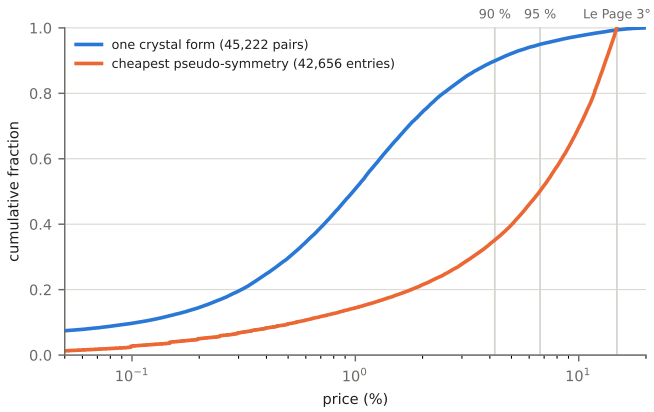


Figure 3
Non-isomorphism against pseudo-symmetry in the PDB: cumulative distributions of the price between deposited cells of one crystal form (blue; one value per pair) and of the price of the cheapest pseudo-symmetric alternative of each entry for which Le Page's 3° finds a higher lattice that is not exact (orange; one value per entry). The orange curve ends near 14.8% by construction, since an obliquity of at most 3° prices a twofold at most so. Vertical lines: the 90th and 95th percentiles of non-isomorphism, 4.2 and 6.7%, and the price of a twofold at Le Page's 3°. Prices are on a logarithmic axis, so pairs of price zero (identical cells, and most cubic pairs) lie off it; the per-type percentiles are in Table 10.

6. Discussion

Completeness. The result of this paper is a statement of when a candidate set is complete. The operators between reduced cells of any lattice lie in one finite set; for a measured cell, the reduced cells within a tolerance are finite and are enumerated exactly; and the candidates within a price band are therefore all found, grouped into classes that differ by pseudo-symmetries whose price is known. The searches that programs use, from unit-cell similarity searches to lattice-symmetry pipelines, find the true operator in ordinary cases; on the cases tested here the $\{-1, 0, 1\}$ window between Niggli cells and the cctbx similarity search found it in every case, and our re-implementation of the lattice-symmetry pipeline missed 19 of 1560 (Section 5.2). The theory says why a fixed list can work: below the adjacency radius every candidate is one of the 6096 matrices of $\mathcal{U}$, so any fixed list that contains their images has the same reach there, and between Niggli cells those images were always within $\{-1, 0, 1\}$; beyond the radius, and wherever such an observation is not enough, the enumeration supplies what a fixed list cannot promise.

A groupoid, and bounds. The exact closure is the part of the action groupoid of $GL_3(\mathbb{Z})$ on metrics that sits on reduced cells (Weinstein, 1996): its loops are point groups, the arrows between two objects form a coset, and all of them are computed once, as $\mathcal{U}$, the neighbours of Voronoi's principal domain (Voronoi, 1908; Schürmann, 2009). A measured cell has no groupoid of its own, only near-symmetries that need not compose within the tolerance; they are the sublevel sets of a displacement, finite because the action is properly discontinuous, and bounded on both sides by the closure of the nearest symmetric lattice (Proposition A10(v)). Thin cells lie far out in the cusps of the space of lattices, where a shear that adds a short vector to a long one moves the metric very little; the number of near-symmetries there has no bound independent of the cell, which is why the ball of Lemma A6 grows for needles and plates at large bands.

Pseudo-symmetry and non-isomorphism. The price of a pseudo-symmetric alternative is a geodesic distance and can be compared directly with the distance between two crystals of one form. Where the alternative costs less than ordinary non-isomorphism, no metric criterion can exclude it and the intensities must decide, as in twinning and indexing-ambiguity tests (Yeates, 1997; Evans, 2006, 2011; Brehm & Diederichs, 2014; Gildea & Winter, 2018); in the PDB that is so for about one entry in twelve. The figure rests on a proxy for one crystal form, adjacent identifiers of one space group, which clustering on the cells or the intensities (Foadi *et al.*, 2013) would sharpen. At about 5% of price per degree of Le Page obliquity, the 90th percentile of non-isomorphism corresponds to about 0.85°, and Le Page's 3°, a tolerance for assigning a Bravais lattice, prices a twofold at two and a half to six times it; for twofolds the price depends on the obliquity and not on the sort, and the sort decides how often each kind of pseudo-symmetry occurs (Table S8).

The Niggli cell. The instability of the Niggli cell (Gruber, 1973; Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same phenomenon in part. At a zero scalar product the Niggli cell changes between type I and type II, and the closure handles that boundary as it handles any zero conorm; at a tie such as $a = b$, and at Niggli's special conditions, the Niggli basis changes while the Selling-reduced superbase does not, being only relabelled (Andrews *et al.*, 2019*a*). Proposition A11 adds the sign condition under which a type I Niggli cell leads to a different Selling-reduced superbase. Tolerant Niggli reduction (Křivý & Gruber, 1976; Grosse-Kunstleve *et al.*, 2004) returns one of these cells, not the set.

Related work. CS6Dist and NCDist are metrics on lattice space, suitable for database search, and return a distance, not an operator. Measuring approximate symmetry by the distance to the symmetric configurations, as Proposition A9 does for lattices, is the idea of continuous symmetry measures for molecules (Zabrodsky *et al.*, 1992). Oishi-Tomiyasu (2012, 2024) bounds the basis transforms needed for Bravais-lattice determination under an error bound; the layers and adjacency radii here do the same for reindexing, with the radius computed for each cell. The same frames give the distance of a cell to each of the 24 sorts, compared with BLDConograph and SELLA in the supporting information (Section S6).

Scope. "Complete" in this paper means: every proper unimodular operator whose price is at most the band is among the candidates, for a band the user supplies. The enumeration is exact below a strain of 25% for every cell whose ball of Lemma A6 the implementation accepts (thin cells at large bands are refused with a message), and the lookup of $\mathcal{U}$ when the widened set lies in the first layer. Sub- and superlattice relations (a doubled axis, determinant ± 2 or more) are outside the closure and are refused with a message; an obverse–reverse relation between two rhombohedral lattices keeps the cell and is

not seen at all; the choice within a class is left to the intensities. The PDB measurements describe deposited cells, refined with their reported symmetry imposed, whose free parameters alone carry error; cells from indexing before symmetry is imposed, or from serial and multi-crystal data, carry error in all six parameters and need wider bands, which a calibration on unconstrained per-crystal cells would give.

7. Conclusions

The reduced cells of a lattice and the integer matrices between them form a groupoid whose arrows, for every lattice, lie in one set of 6096 matrices; that completeness, its converse and the dependence on the zero pattern alone are machine-checked. For a measured cell the reduced cells within a strain tolerance are enumerated exactly, so reindexing returns every proper operator within a stated price band, and prices, as displacements, place the candidates between cosets of the near-symmetries, bounded by the closure of the nearest symmetric lattice. In practice (Table 7): widen each reference once at the chosen band, Selling-reduce each new data set once, group the candidates by the Laue group, and read each class's price against the non-isomorphism of its Bravais type (Table 10). The default band of 7.4% lies above the 95th percentile of non-isomorphism, so about 4% of real pairs of one crystal form differ by more than the band, and for them the true operator is not returned; the band is a choice of percentile, not a bound; a band of 9.9 to 14.8% reproduces the alternatives that lattice-symmetry searches test, and within the band the intensities decide.

Data availability. The implementation is the module `agentsg.cell.strain_closure` of `agentsg` (Zwart, 2026; `github.com/phzwart/agentsg`), version 0.5.0. Its main functions are `widened_closure`, `reindex`, `indexing_classes` and `universal_operators`; the 6096 column operators of $\mathcal{U}$ are also deposited as `manuscript/benchmark/universal_operators_columns.json`. The PDB cell snapshot is on Zenodo (doi:10.5281/zenodo.22986222). The Lean proofs are in `manuscript/formal`. An archived release of the repository, with a DOI, will accompany publication [DOI to be inserted]. Every number in this paper is reproduced by a test in `manuscript/appendix_z`, by `agentsg/tests/test_strain_closure.py` or by a script in `manuscript/benchmark`; the supporting information lists them by section, table and figure (Table S12), with the versions of the software used.

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Appendix A
Proofs

Terms. Two lattice vectors are in the same *parity class* when their difference is twice a lattice vector; the classes are the eight cosets of 2Λ in Λ, one of them 2Λ itself, and the other seven are used here. In a basis v_1, v_2, v_3 every lattice vector is $x = n_1v_1 + n_2v_2 + n_3v_3$ with integer coefficients n_i, and $x - y$ is twice a lattice vector exactly when every coefficient of $x - y$ is even. So a vector's class is read from the parities of its coefficients, the pattern $(n_1, n_2, n_3) \bmod 2$, with $(0,0,0)$ for 2Λ. The pattern is a label that depends on the basis: a unimodular change of basis maps the coefficients by an integer matrix whose inverse is also integer, which keeps even coefficients even, so it permutes the labels and leaves the classes unchanged. For the superbase (v_0, v_1, v_2, v_3), $v_0 = -v_1 - v_2 - v_3$, a vector $x = \sum_{i=0}^{3} c_i v_i$ has integer coefficients c_i determined only up to adding a common integer; adding an odd one complements their parity pattern, so a pattern $s \in \{0,1\}^4$ and $1 - s$ name the same class, and we fix $c_0 = 0$. A *partial sum* of a superbase is the sum of some of its four vectors; there are seven up to sign, one in each of the seven classes. A superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, that is, a Voronoi vector of the lattice (Voronoi, 1908; Conway & Sloane, 1992; Kurlin, 2022); at a zero conorm the partial sums are Voronoi vectors that are not strict. The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum in that class minus the squared length μ_c of the shortest vector of the class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$; a vector of the same class has coefficients $s+2k$ with k integer. Every lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{6}$$

whatever the signs of the conorms, because the four vectors sum to zero.

Lemma A1 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial sum and w' a vector of its parity class with coefficients $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + \Delta s)$, $\Delta k = k_i - k_j$, $\Delta s = s_i - s_j$; each γ_{ij} is a non-negative integer, zero exactly when $\Delta k \in \{0, -\Delta s\}$. *Proof.* Equation (6), $(\Delta s+2\Delta k)^2 - \Delta s^2 = 4\Delta k(\Delta k+\Delta s)$, and $t(t+u) \geq 0$ for integers t and $u \in \{-1, 0, 1\}$. □

Lemma A2 (window). Let F_0 and F be obtuse superbases of the same lattice. Then every vector of F has superbase coefficients $s_i + 2k_i$ relative to F_0, with $s \in \{0,1\}^4$, $k_0 = 0$ and $|k_i| \leq 3$; the coefficients are defined up to a common integer, which $k_0 = 0$ fixes (in the basis (a, b, c) they then lie in $[-7, 7]$, the window of the machine-checked version). *Proof.* The non-zero conorms of F_0 connect its four vectors, since otherwise a partial sum would have length zero by equation (6). Every partial sum of F is a shortest vector of its parity class, as is the partial sum of F_0 in that class, so their excess vanishes; by Lemma A1 the excess is $4\sum p^0_{ij}\gamma_{ij}$ with every term non-negative, so $\gamma_{ij} = 0$ on every non-zero slot, i.e. $|k_i - k_j| \leq 1$ along a connected graph on four vertices, and with $k_0 = 0$ every $|k_i| \leq 3$. □ Enumerating, over all 38 faces, the superbases of such vectors whose six pairwise prod-

ucts are non-positive on the whole face gives 1, 2, 3, 6 and 16 superbases for no zero, one zero, two disjoint zeros, two zeros sharing a vector and three zeros (both placements), independently of Kurlin's lists. In the basis of F_0, over every labelling of each member, their entries are in $\{-1,0,1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 4): the conorms $(0,0,3,2,0,5)$ have the obtuse basis $(-3a-2b-c,\ a,\ a+b)$. Every member lies in $\mathcal{U}$ in each of its 48 labellings, which is the computer-assisted proof that $\mathcal{U}$ contains every arrow, and every member has a labelling with entries in $[-2,2]$. Relabelling F_0 once for all members, the largest entry is 1 on every face but the path of three zeros, where it is 2; relabelling both F_0 and each member, every entry is in $\{-1,0,1\}$.

Lemma A3 (generation). Every arrow of the closure, and every symmetry, is a word of flips at zero conorms followed by a swap. *Proof.* Let F and F' be obtuse superbases of a lattice with metric $\mathbf{G}$. Add a small positive amount, in the frame of F, to each conorm of F that vanishes under $\mathbf{G}$; this gives a metric $\mathbf{G}'$ for which F is strictly obtuse and hence the only obtuse superbase, and, the lattice having finitely many obtuse superbases, the amount can be chosen so small that every conorm that is positive under $\mathbf{G}$ for any of them stays positive under $\mathbf{G}'$. Selling reduction under $\mathbf{G}'$ started from F' flips only at negative conorms. The first such conorm is zero under $\mathbf{G}$, so the flip carries F' to another obtuse superbase of $\mathbf{G}$, and the same holds at every later step. Reduction terminates (Selling, 1874; Conway & Sloane, 1992), at the only obtuse superbase of $\mathbf{G}'$, which is F. So F' is joined to F by flips at zero conorms of $\mathbf{G}$, and a swap matches the labellings. A symmetry maps F_0 to an obtuse superbase of the lattice; as a matrix it is that flip word followed by a swap.□ Kurlin (2022) lists the superbases of each Voronoi type, and the enumeration of Lemma A2 reproduces the counts.

Proposition A4 (the strain norm bounds every dot product). If $\|\mathbf{G}^{-1/2}\boldsymbol{\Delta}\mathbf{G}\,\mathbf{G}^{-1/2}\|_F \le \boldsymbol{\eta}$, then $|u^{\mathsf{T}}\boldsymbol{\Delta}\mathbf{G}\,v| \le \boldsymbol{\eta}\,|u|\,|v|$ for all vectors u, v, lengths in $\mathbf{G}$; in particular every conorm of every superbase changes by at most $\eta|u_i||u_j|$. *Proof.* With $X = \mathbf{G}^{-1/2}\boldsymbol{\Delta}\mathbf{G}\,\mathbf{G}^{-1/2}$, $u^{\mathsf{T}}\boldsymbol{\Delta}\mathbf{G}\,v = (\mathbf{G}^{1/2}u)^{\mathsf{T}}X(\mathbf{G}^{1/2}v)$, at most $\|X\|_2|u||v| \le \|X\|_F|u||v|$. □

Definition and enumeration A5 (sorts as closed sets of equalities). Fix a zero pattern Z, let O_Z be the arrows of its closure and, for $\mathbf{W} \in O_Z$, let $\mathbf{A}_\mathbf{W}$ be its linear action on conorm vectors (supporting information, Section S3), which on the face $\{p \ge 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace V of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : \mathbf{A}_\mathbf{W} = \mathrm{id} \text{ on } V\}$ and $\mathrm{Fix}(H) = \{p : \mathbf{A}_\mathbf{W} p = p \ \forall\, \mathbf{W} \in H\}$. A lattice with conorm vector p on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in O_Z by Lemma A3. We take as the definition of a sort a maximal set of lattices with the same kind of closure and the same symmetry operators; then the sorts of kind Z are the generic points of the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the arrows that map the face onto itself. This is the classical construction, the fixed subspace of a point group in the space of forms (Delone *et al.*, 1975), written in conorm coordinates; SELLA uses the same subspaces. Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$; since $\mathbf{A}_\mathbf{W}$ permutes the slots on the face, each is a set of equalities among conorms. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three zeros, with symmetry orders $2,4,4,8,8,12,16,48$; $2,4,4,8,16$; $4,8,24$; $2,4,8,12,48$; $8,16,48$. A generic point of a closed subspace, one in the interior of the face and in none of the finitely many smaller closed subspaces it contains, has exactly the symmetries $\mathrm{Sym}(V)$; points on its boundary belong to another zero pattern. Matched pattern by pattern, by computation, these are Delone's 24 sorts (supporting information, Table S10).

Lemma A6 (the widened set is finite). If F is within strain $\eta < 1$ of obtuse for $\mathbf{G}$, every vector u of F satisfies $|u|^2_\mathbf{G} \le \frac{1+\eta}{1-\eta}\,\mu_c(\mathbf{G})$, with μ_c the squared length of the shortest vector of the parity class of u, which is that of F_0's partial sum in the class. *Proof.* Let $\mathbf{G}'$ within η make F obtuse. By Proposition A4 with $u = v$, $(1-\eta)|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'} \le (1+\eta)|u|^2_\mathbf{G}$ for every lattice vector. The vectors of F are shortest in their classes for $\mathbf{G}'$, and those of F_0 for $\mathbf{G}$, so $|u|^2_{\mathbf{G}'} \le |v^0_c|^2_{\mathbf{G}'} \le (1+\eta)\mu_c(\mathbf{G})$ and $|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'}/(1-\eta)$. □ The lattice vectors in these seven balls are finitely many and computable from $\mathbf{G}$, so the candidates for $\mathcal{S}_\eta(\mathbf{G})$ are enumerated exactly.

Lemma A7 (widening one side suffices). (i) If $D(\mathbf{A},\mathbf{B}) \le \boldsymbol{\epsilon}$, then $\|\mathbf{B}-\mathbf{A}\|_\mathbf{A} \le e^{\boldsymbol{\epsilon}} - 1$. (ii) Let $\mathbf{P}$ be an operator from a reference with metric $\mathbf{G}_R$ to a new cell with metric $\mathbf{G}_N$, of price at most ϵ, and let F be obtuse for $\mathbf{G}_N$. Then $\mathbf{P}F \in \mathcal{S}_{e^\epsilon-1}(\mathbf{G}_R)$. *Proof.* (i) The eigenvalues of $\mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$ are e^{l_i} with $\sum l_i^2 \le \epsilon^2$, and $|e^l-1| \le |l|(e^a-1)/a$ for $|l| \le a$, because $(e^x-1)/x$ increases; with $a = \max|l_i| \le \epsilon$, $\|\mathbf{B}-\mathbf{A}\|_\mathbf{A} = (\sum(e^{l_i}-1)^2)^{1/2} \le \boldsymbol{\epsilon}\,(e^{\boldsymbol{\epsilon}}-1)/\boldsymbol{\epsilon}$. (ii) The price is $D(\mathbf{G}_N, s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P})$ for the fitted scale s. Carried back by $\mathbf{P}$, an isometry of both measures, $\mathbf{P}F$ is obtuse for $\mathbf{M} = \mathbf{P}^{-\mathsf{T}}\mathbf{G}_N\mathbf{P}^{-1}$, and $D(s\mathbf{G}_R,\mathbf{M}) \le \boldsymbol{\epsilon}$, so by (i) $\mathbf{M}$ is within strain $e^\epsilon - 1$ of $s\mathbf{G}_R$; the strain and the widened set are unchanged by the scale. □ For the price band of 7.4% the reference is widened at 7.7%.

Remark A8 (layers, and the lookup of $\mathcal{U}$). Two superbases are *adjacent* when some positive definite metric makes both obtuse; a superbase adjacent to F_0 is $\mathbf{W}F_0$ with $\mathbf{W} \in \mathcal{U}$, since both are superbases of that metric's closure. Let $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$, and define $r_k(\mathbf{G})$ as the smallest strain from $\mathbf{G}$ to a metric for which a superbase outside $\mathcal{N}_k$ is obtuse; so if $\eta < r_k(\mathbf{G})$, $\mathcal{S}_\eta(\mathbf{G})$ is the set of members of $\mathcal{N}_k$ within strain η of obtuse. The content of this definition is the description of $\mathcal{N}_1$, which rests on the completeness of $\mathcal{U}$. *The lookup, a corollary of Lemma A7 and the completeness of $\mathcal{U}$.* Let $\mathbf{P}$ be an operator of price at most ϵ from a reference with reduced basis B_R to a new cell with reduced basis B_N, and let the reference's widened set at $e^\epsilon - 1$ lie in $\mathcal{N}_1$, as it does when $e^\epsilon - 1 < r_1$. By Lemma A7 the new cell's reduced superbase, carried back by $\mathbf{P}$, lies in $\mathcal{S}_{e^\epsilon-1}(\mathbf{G}_R) \subseteq \mathcal{N}_1$, so it is $\mathbf{W}F_0$ for some $\mathbf{W} \in \mathcal{U}$ in some labelling, and $\mathbf{P} = B_R\mathbf{W}\mathbf{Q}B_N^{-1}$ with $\mathbf{Q}$ a swap. Since $\mathcal{U}$ is closed under swaps, $\mathbf{W}\mathbf{Q} \in \mathcal{U}$: trying $B_R\mathbf{W}B_N^{-1}$ for the 6096 column operators $\mathbf{W}$, or for the 3048 with $\det\mathbf{W} = \det B_R \det B_N$ when $\mathbf{P}$ is proper, finds $\mathbf{P}$. □ The radii are computed from the balls of Lemma A6 at a strain of $\frac{1}{4}$, with the depths from a floating-point convex solver, and those of Table 8

are certified in exact arithmetic by a dual bound and a feasible point for each depth (supporting information, Section S7). No lower bound on r_1 in terms of the cell's geometry is proved; for the cells of Table 8 below the cap it is of the order of the ratio of the shortest to the longest edge.

Proposition A9 (the price of a symmetry). Let g be an integer matrix of finite order n and $V_g = \{\mathbf{G} : g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}\}$ the metrics for which g is a symmetry. (i) $D(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) \le 2D(\mathbf{G}, V_g)$, with equality when g is a twofold. (ii) If g is a twofold along a lattice row of obliquity δ in $\mathbf{G}$ (the angle between the row and the normal to the lattice plane that g negates), then $D(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) = 2\sqrt{2}\,\operatorname{artanh}(\sin\boldsymbol{\delta})$. (iii) For a metric $\mathbf{G}^* \in V_g$ and the strain at $\mathbf{G}^*$, $\|g^{\mathsf{T}}\mathbf{G}g - \mathbf{G}\|_{\mathbf{G}^*}$ lies between $\rho_{\min}$ and $\rho_{\max}$ times the distance from $\mathbf{G}$ to V_g in that norm, with $\rho = |e^{i\theta} - 1|$ over the rotation angles $\theta \neq 0$ of $X \mapsto g^{\mathsf{T}}Xg$: $\rho = 2$ for a twofold, $\sqrt{3}$ for a threefold, $\{\sqrt{2}, 2\}$ for a fourfold and $\{1, \sqrt{3}\}$ for a sixfold. The scale fit leaves these prices unchanged, since $\det g^{\mathsf{T}}\mathbf{G}g = \det\mathbf{G}$. *Proof.* (i) $\mathbf{G} \mapsto g^{\mathsf{T}}\mathbf{G}g$ is an isometry of D. For any $\mathbf{G}' \in V_g$, $D(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) \le D(\mathbf{G}, \mathbf{G}') + D(\mathbf{G}', g^{\mathsf{T}}\mathbf{G}g) = 2D(\mathbf{G}, \mathbf{G}')$. For a twofold, the geodesic midpoint m of $\mathbf{G}$ and $g^{\mathsf{T}}\mathbf{G}g$ is unique, the space of positive definite matrices with this distance being complete and of non-positive curvature (Bhatia, 2007, Chapter 6), and g exchanges the two ends, so $m \in V_g$ and $D(\mathbf{G}, V_g) \le \frac{1}{2}D(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g)$. (ii) In Cartesian coordinates for $\mathbf{G}$, g becomes $T = \mathbf{G}^{1/2}g\mathbf{G}^{-1/2}$, an involution fixing the line ℓ of the row and negating the plane Π; it is $2\mathbf{E} - I$ with $\mathbf{E}$ the projection onto ℓ along Π, and the angle between ℓ and the normal of Π is δ. The price is $\|\ln(T^{\mathsf{T}}T)\|_F$. $T^{\mathsf{T}}T$ is the identity on the line of Π normal to ℓ, and on the plane spanned by ℓ and the normal of Π it has eigenvalues $(1 \pm \sin\delta)/(1 \mp \sin\delta)$, whose logarithms are $\pm 2\operatorname{artanh}(\sin\delta)$. (iii) $\sigma : X \mapsto g^{\mathsf{T}}Xg$ is an isometry of the inner product $\operatorname{tr}(\mathbf{G}^{*-1}X\mathbf{G}^{*-1}Y)$ of order n, so the space of symmetric matrices splits orthogonally into its fixed space, which is $V_g - \mathbf{G}^*$, and planes and lines on which it rotates by $\theta = 2\pi k/n$. With $\mathbf{G} - \mathbf{G}^* = X_0 + X_\perp$, X_0 fixed, the distance to V_g is $\|X_\perp\|$ and $g^{\mathsf{T}}\mathbf{G}g - \mathbf{G} = (\boldsymbol{\sigma} - 1)X_\perp$. The rotation angles are those of the products $\lambda_i\lambda_j$, $i \le j$, of the eigenvalues of g, squares included; the squares give the angle π of a fourfold and $2\pi/3$ of a sixfold. □ Part (iii) is exact for the strain at $\mathbf{G}^*$; for the price it holds to first order in the distance to V_g. No exact relation to Le Page's obliquity holds for orders 3, 4 and 6: at Le Page's boundary the fourfolds of a body-centred cubic lattice cost a median 15.0%, more than its twofolds (supporting information, Table S11). The metrics of a sort to which g belongs lie inside V_g (read in the frame in which the sort's distance is attained), so, to first order in the error, where the strain of a sort distance and the geodesic distance agree, a twofold's price is at most twice the distance to that sort, with equality when V_g is the sort's subspace, as for the twofold of a primitive monoclinic lattice.

Proposition A10 (candidates and cosets). Let $\mathbf{P}_0$ be the true operator from a reference R to a new cell N, of price p_0; let H_ϵ be the proper unimodular h with $\pi(h) = D(\mathbf{G}_R, h^{\mathsf{T}}\mathbf{G}_R h) \le \boldsymbol{\epsilon}$, a finite set, since $\pi(h) \le \epsilon$ bounds the eigenvalues of $\mathbf{G}_R^{-1}h^{\mathsf{T}}\mathbf{G}_R h$ by e^{ϵ}, so that every column of h has $|he_i|^2 \le e^{\epsilon}|e_i|^2$; and let C_ϵ be the candidates, the proper unimodular operators of price at most ϵ. Then (i) $\pi(h_1h_2) \le \pi(h_1) + \pi(h_2)$; (ii) $H_{\epsilon-p_0}\mathbf{P}_0 \subseteq C_{\boldsymbol{\epsilon}} \subseteq H_{\boldsymbol{\epsilon}+p_0}\mathbf{P}_0$; (iii) if no self-map of R has price in $(\epsilon, 2\epsilon]$, H_ϵ is a group; (iv) if none has price in $(\epsilon - p_0, \epsilon + p_0]$, then $C_\epsilon = H_\epsilon\mathbf{P}_0$; (v) if $\mathbf{G}_0$ is any metric at distance ν from $\mathbf{G}_R$, after normalisation to unit determinant, and $\operatorname{Aut}(\mathbf{G}_0)$ its point group, then $\operatorname{Aut}(\mathbf{G}_0) \subseteq H^{\pm}_{2\boldsymbol{\nu}}(\mathbf{G}_R)$ and $H^{\pm}_{\epsilon}(\mathbf{G}_R) \subseteq H^{\pm}_{\boldsymbol{\epsilon}+2\boldsymbol{\nu}}(\mathbf{G}_0)$, where $H^{\pm}$ admits improper self-maps; in particular, if $\mathbf{G}_0$ has no self-map outside its point group of price at most $\epsilon + 2\nu$, and $\epsilon \ge 2\nu$, then $H^{\pm}_{\epsilon}(\mathbf{G}_R) = \operatorname{Aut}(\mathbf{G}_0)$. *Proof.* The price is the geodesic distance between metrics normalised to unit determinant, and unimodular matrices act on those by isometries. (i)

$$D(\mathbf{G}_R, (h_1h_2)^{\mathsf{T}}\mathbf{G}_R h_1h_2) \le D(\mathbf{G}_R, h_2^{\mathsf{T}}\mathbf{G}_R h_2) + D(h_2^{\mathsf{T}}\mathbf{G}_R h_2, h_2^{\mathsf{T}}h_1^{\mathsf{T}}\mathbf{G}_R h_1h_2) = \boldsymbol{\pi}(h_2) + \boldsymbol{\pi}(h_1).$$

(ii) By the triangle inequality the price of $h\mathbf{P}_0$ differs from $D(\mathbf{P}_0^{\mathsf{T}}\mathbf{G}_R\mathbf{P}_0, \mathbf{P}_0^{\mathsf{T}}h^{\mathsf{T}}\mathbf{G}_R h\mathbf{P}_0) = \boldsymbol{\pi}(h)$ by at most p_0, and every unimodular $\mathbf{P}$ is $h\mathbf{P}_0$ with $h = \mathbf{P}\mathbf{P}_0^{-1}$. (iii) By (i) a product of two elements of H_ϵ has price at most 2ϵ, hence at most ϵ; a finite set of matrices closed under products is a group. (iv) The two sets of (ii) then coincide. (v) For $g \in \operatorname{Aut}(\mathbf{G}_0)$,

$$D(\mathbf{G}_R, g^{\mathsf{T}}\mathbf{G}_R g) \le D(\mathbf{G}_R, \mathbf{G}_0) + D(\mathbf{G}_0, g^{\mathsf{T}}\mathbf{G}_0 g) + D(g^{\mathsf{T}}\mathbf{G}_0 g, g^{\mathsf{T}}\mathbf{G}_R g) = 2\boldsymbol{\nu};$$

for $h \in H^{\pm}_{\epsilon}(\mathbf{G}_R)$, in the same way, $D(\mathbf{G}_0, h^{\mathsf{T}}\mathbf{G}_0 h) \le \boldsymbol{\nu} + \boldsymbol{\epsilon} + \boldsymbol{\nu}$. The last statement combines the two inclusions. □
The conditions are sufficient, not necessary: the worked cell at $\epsilon = 7.4\%$ has self-maps of price 0, 1.55% and 10.2%, so the condition of (iii) fails, yet $H_\epsilon = \{\mathbf{I}, 2_y\}$ is a group, which the implementation checks directly.

Proposition A11 (Niggli type and the zeros on d). Let (a, b, c) be a Niggli cell, $a \cdot a \le b \cdot b \le c \cdot c$, with Niggli's conditions $|2a \cdot b| \le a \cdot a$, $|2a \cdot c| \le a \cdot a$, $|2b \cdot c| \le b \cdot b$. (i) If it is of type II ($a \cdot b$, $a \cdot c$, $b \cdot c \le 0$), the superbase (d, a, b, c) is obtuse (a known relation between the two reductions; Andrews *et al.*, 2019*a*), and the conorm of d with each vector x is at least $\frac{1}{2}x \cdot x$ minus the smaller of $|x \cdot y|$ and $|x \cdot z|$; so if at most one of $|a \cdot b|$, $|a \cdot c|$, $|b \cdot c|$ exceeds τ and $\tau < \frac{1}{4}a \cdot a$, every conorm on d exceeds every conorm off d that is at most τ. (ii) If it is of type I (all three products positive), none of the four orthogonal superbases $\{\pm a, \pm b, \pm c,\ \mp a \mp b \mp c\}$ is obtuse. (iii) Let $\mathbf{G}^*$ be a lattice with three mutually perpendicular shortest vectors e_1, e_2, e_3, and $\mathbf{G}$ a metric within strain η of it, small enough that every superbase obtuse for $\mathbf{G}$ belongs to the closure of $\mathbf{G}^*$; the largest such η is the strain at which a superbase outside that closure first becomes obtuse near $\mathbf{G}^*$, which the ball enumeration of Lemma A6 computes. If $(e_1 \cdot e_2)(e_1 \cdot e_3)(e_2 \cdot e_3) > 0$ under $\mathbf{G}$, which for the near-orthogonal Niggli cell $(\pm e_1, \pm e_2, \pm e_3)$ is Niggli's type I, the reduced superbase of $\mathbf{G}$, from any starting basis, is a face-diagonal cell, whose three zeros under $\mathbf{G}^*$ form a path through d. *Proof.* (i) p_{ab}, p_{ac}, $p_{bc} \ge 0$ by the signs; $p_{dx} = x \cdot x + \sum_{y \neq x} x \cdot y \ge x \cdot x - |x \cdot y| - |x \cdot z|$, and the larger of the two is at most $\frac{1}{2}x \cdot x$ by Niggli's conditions and the ordering of the lengths, which gives the bound; if at most one product exceeds τ, the smaller product of x is at most τ,

so $p_{dx} \geq \frac{1}{2}x \cdot x - \tau > \tau$ when $\tau < \frac{1}{4}a \cdot a \leq \frac{1}{4}x \cdot x$. (ii) An orthogonal superbase is obtuse only if its three basis products are non-positive, so their product is non-positive; changing the signs of basis vectors leaves $(a \cdot b)(a \cdot c)(b \cdot c)$ unchanged, and it is positive. (iii) A superbase obtuse for $\mathbf{G}$ is in the closure of $\mathbf{G}^*$ by hypothesis. That closure has sixteen superbases, the four orthogonal ones, whose zeros form a triangle, and twelve face-diagonal ones, whose zeros form a path that touches every vector and so d in every labelling; by (ii), applied to the orthogonal triple, only the latter can be obtuse. □

References

Andrews, L. C. & Bernstein, H. J. (1988). *Acta Cryst.* A**44**, 1009–1018.
Andrews, L. C. & Bernstein, H. J. (2014). *J. Appl. Cryst.* **47**, 346–359.
Andrews, L. C. & Bernstein, H. J. (2023). *Acta Cryst.* A**79**, 499–503.
Andrews, L. C., Bernstein, H. J. & Pelletier, G. A. (1980). *Acta Cryst.* A**36**, 248–252.
Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2019a). *Acta Cryst.* A**75**, 115–120.
Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2019b). *Acta Cryst.* A**75**, 593–599.
Andrews, L. C., Bernstein, H. J. & Sauter, N. K. (2023). SELLA – a program for determining Bravais lattice types. arXiv:2303.03122.
Bhatia, R. (2007). *Positive Definite Matrices.* Princeton: Princeton University Press.
Brandt, H. (1927). *Math. Ann.* **96**, 360–366.
Brehm, W. & Diederichs, K. (2014). *Acta Cryst.* D**70**, 101–109.
Bridson, M. R. & Haefliger, A. (1999). *Metric Spaces of Non-Positive Curvature.* Berlin: Springer.
Burzlaff, H. & Zimmermann, H. (1985). *Z. Kristallogr.* **170**, 247–262.
Burzlaff, H. & Zimmermann, H. (2006). *International Tables for Crystallography*, Vol. A, 5th ed., edited by Th. Hahn, ch. 9.1, pp. 742–749. Dordrecht: Springer.
Conway, J. H. & Sloane, N. J. A. (1992). *Proc. R. Soc. London Ser. A*, **436**, 55–68.
Dauter, Z. (2003). *Acta Cryst.* D**59**, 2004–2016.
Delaunay, B. (1933). *Z. Kristallogr.* **84**, 109–149.
Delone, B. N., Galiulin, R. V. & Shtogrin, M. I. (1975). *J. Sov. Math.* **4**, 79–156.
de Moura, L. & Ullrich, S. (2021). *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, Vol. 12699, pp. 625–635. Cham: Springer.
de Wolff, P. M. & Gruber, B. (1991). *Acta Cryst.* A**47**, 29–36.
Evans, P. R. (2006). *Acta Cryst.* D**62**, 72–82.
Evans, P. R. (2011). *Acta Cryst.* D**67**, 282–292.
Flack, H. D. (1987). *Acta Cryst.* A**43**, 564–568.
Foadi, J., Aller, P., Alguel, Y., Cameron, A., Axford, D., Owen, R. L., Armour, W., Waterman, D. G., Iwata, S. & Evans, G. (2013). *Acta Cryst.* D**69**, 1617–1632.
Gildea, R. J. & Winter, G. (2018). *Acta Cryst.* D**74**, 405–410.
Grosse-Kunstleve, R. W., Sauter, N. K. & Adams, P. D. (2004). *Acta Cryst.* A**60**, 1–6.
Grosse-Kunstleve, R. W., Sauter, N. K., Moriarty, N. W. & Adams, P. D. (2002). *J. Appl. Cryst.* **35**, 126–136.
Gruber, B. (1973). *Acta Cryst.* A**29**, 433–440.
Hahn, Th. (2006). Editor. *International Tables for Crystallography*, Vol. A, 5th ed. Dordrecht: Springer.
Higgins, P. J. (1971). *Notes on Categories and Groupoids.* London: Van Nostrand Reinhold.
Himes, V. L. & Mighell, A. D. (1987). *Acta Cryst.* A**43**, 375–384.
Křivý, I. & Gruber, B. (1976). *Acta Cryst.* A**32**, 297–298.
Kurlin, V. (2022). *A complete isometry classification of 3-dimensional lattices.* arXiv:2201.10543.
Le Page, Y. (1982). *J. Appl. Cryst.* **15**, 255–259.
mathlib Community (2020). *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 367–381. New York: ACM.
Niggli, P. (1928). *Kristallographische und strukturtheoretische Grundbegriffe.* Handbuch der Experimentalphysik, Vol. 7, Part 1. Leipzig: Akademische Verlagsgesellschaft.
Oishi-Tomiyasu, R. (2012). *Acta Cryst.* A**68**, 525–535.
Oishi-Tomiyasu, R. (2024). *Acta Cryst.* A**80**, 339–350.
Pennec, X., Fillard, P. & Ayache, N. (2006). *Int. J. Comput. Vis.* **66**, 41–66.
Sauter, N. K., Grosse-Kunstleve, R. W. & Adams, P. D. (2004). *J. Appl. Cryst.* **37**, 399–409.
Schürmann, A. (2009). *Computational Geometry of Positive Definite Quadratic Forms.* University Lecture Series, Vol. 48. Providence: American Mathematical Society.
Selling, E. (1874). *J. Reine Angew. Math.* **77**, 143–229.
Voronoi, G. (1908). *J. Reine Angew. Math.* **134**, 198–287.
Weinstein, A. (1996). *Notices Am. Math. Soc.* **43**, 744–752.
Yeates, T. O. (1997). *Methods Enzymol.* **276**, 344–358.
Zabrodsky, H., Peleg, S. & Avnir, D. (1992). *J. Am. Chem. Soc.* **114**, 7843–7851.
Zwart, P. H. (2026). `agentsg`: a space-group and lattice engine, version 0.5.0. https://github.com/phzwart/agentsg.